

Marcus Vance

Montreal, QC, Canada | (514) 555-0183 | marcus.vance.dev@email.com
linkedin.com/in/marcus-vance-backend | github.com/mvance-dev

PROFESSIONAL SUMMARY

Senior Backend Engineer with over 7 years of experience specializing in high-throughput distributed systems, microservices architecture, and API design. Proven expertise in building asynchronous backend engines using Python (FastAPI, Asyncio, Django) and Go, backed by robust database optimization in PostgreSQL and distributed caching with Redis. Skilled in designing fault-tolerant, event-driven architectures with Kafka and RabbitMQ, with a strong focus on system performance, security, and cloud scalability.

CORE COMPETENCIES & SKILLS

Backend Languages: Python (FastAPI, Asyncio, Django, Pydantic), Go, SQL (Advanced), Bash

APIs & Architecture: RESTful APIs, gRPC, GraphQL, Event-Driven Microservices, WebSockets

Databases & Caching: PostgreSQL (Query Tuning, Indexing, Partitioning), Redis, MongoDB, Elasticsearch

Messaging & Streaming: Apache Kafka, RabbitMQ, AWS SQS/SNS

Cloud & Infrastructure: AWS (ECS, S3, RDS, Lambda), Docker, Kubernetes, Terraform, CI/CD

PROFESSIONAL EXPERIENCE

Senior Backend Engineer

Montreal, QC

Apex Distributed Systems

Jan 2023 – Present

- Architected an asynchronous API gateway using **FastAPI** and **Pydantic V2**, handling over 15,000 requests per second with sub-20ms p99 response times.
- Redesigned core database schema in **PostgreSQL**, implementing connection pooling and index optimization that reduced database CPU utilization by 45% during peak loads.
- Implemented a distributed caching and rate-limiting tier using **Redis** and **Lua scripts**, preventing system degradation during traffic surges.
- Mentored 4 backend developers in writing clean, modular Python code and established strict unit and integration testing pipelines with **Pytest** (achieving 92% code coverage).

Backend Software Engineer

Toronto, ON

CloudPulse Solutions

May 2019 – Dec 2022

- Engineered event-driven background processing pipelines utilizing **RabbitMQ** and **Celery**, processing over 4 million background tasks daily.
- Migrated a monolithic Django application into containerized **Go** and **FastAPI** microservices deployed on **AWS ECS** via **Terraform**.
- Designed secure OAuth2 / JWT authentication modules and role-based access control (RBAC) middleware for multi-tenant enterprise clients.

- Collaborated with frontend engineering teams to establish strict JSON schemas and API contracts, reducing cross-team integration friction by 50%.

PROJECTS

High-Speed Async Task Queue Engine

github.com/mvance-dev/async-task-engine

- Built an open-source lightweight task queue in Python using Asyncio and Redis streams, capable of processing 50,000 tasks/min with zero external dependencies.

EDUCATION

Bachelor of Science in Computer Science

Concordia University

Montreal, QC

Sep 2015 – Apr 2019

CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate** | Amazon Web Services (Issued: Oct 2023)
- **Certified Kubernetes Application Developer (CKAD)** | Cloud Native Computing Foundation (Issued: May 2022)